The Nurse Scheduling Problem

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1 Description of the Problem

The Nurse Scheduling Problem (NSP) involves creating an optimal shift schedule for nurses working in a hospital over a given period of days, subject to some constraints:

(1) Each shift requires a minimum number of nurses for the normal functioning of the hospital.
(2) Each shift must have at least one nurse with managerial qualifications.
(3) A nurse's assigned shifts on any given day must be continuous, with no gaps in between.
(4) Each nurse has a minimum and maximum daily workload, defined by the number of assigned shifts.
(5) The total number of shifts that can be assigned to a nurse for the entire period is limited.
(6) A nurse cannot be assigned to shifts when it's known they're unavailable to work.
(7) The schedule must consider nurses' shift preferences. An optimal solution aims to minimize the total cost associated with the assignments.

2 Description of the Encoding

This problem can be modeled and solved using Maximum Satisfiability (MaxSAT). We first define a set of variables to represent the schedule and then encode the problem's constraints as clauses. This encoding is then passed to a MaxSAT solver to find an optimal solution.

2.1 Definition of Variables

- Let D be a positive integer representing the number of days in the scheduling period.
- Let S be a positive integer representing the number of shifts per day, all with the same duration, that cover the full 24 hours. For example, if $S = 4$, each shift will have 6 hours.
- Let $R_{s,d}$ be the minimum number of nurses required for shift s ($1 \leq s \leq S$) on day d ($1 \leq d \leq D$).
- Let $N = \{N_1, N_2, \ldots, N_n\}$ be the set of n available nurses.
- Let $M \subseteq N$ be the subset of nurses with managerial qualifications.
- Let $k_{\min}$ and $k_{\max}$ be the minimum and maximum number of shifts for nurse N_k ($1 \leq k \leq n$) in any given day.
- Let H_k be the maximum total number of shifts that can be assigned to nurse N_k over the entire period.
- Let $P_{k,s,d}$ be the cost of assigning nurse N_k to shift s on day d. If $P_{k,s,d} = -1$, nurse N_k is not available for that shift. All other costs are positive integers representing preferences (a lower cost is better).
- Let $x_{k,s,d}$ be a Boolean variable such that $x_{k,s,d} = 1$ if nurse N_k is assigned to shift s on day d, and $x_{k,s,d} = 0$ otherwise.

2.2 Modeling Constraints

The constraints are modeled as a combination of hard and soft clauses. Hard clauses must be satisfied, while soft clauses represent preferences that the solver will try to satisfy by minimizing the total weight of the unsatisfied clauses.

2.2.1 *Hard Clauses.*

(1) **Minimum nurses per shift:**

$$\sum_{k=1}^{n} x_{k,s,d} \geq R_{s,d}, \quad \forall s \in [1, S], d \in [1, D] \tag{1}$$

(2) **Manager requirement:**

$$\sum_{N_k \in M} x_{k,s,d} \geq 1, \quad \forall s \in [1, S], d \in [1, D] \tag{2}$$

(3) **Shift continuity:**

$$\begin{gathered}(\neg x_{k,s,d} \vee x_{k,s+1,d} \vee \neg x_{k,v,d}), \\ \forall N_k \in N, d \in [1, D], s \in [1, S-1], v \in [s+2, S]\end{gathered} \tag{3}$$

(4) **Daily workload:**

$$k_{\min} \leq \sum_{s=1}^{S} x_{k,s,d} \leq k_{\max}, \quad \forall N_k \in N, d \in [1, D] \tag{4}$$

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(5) **Total workload:**

$$\sum_{d=1}^{D} \sum_{s=1}^{S} x_{k,s,d} \leq H_k, \quad \forall N_k \in N \tag{5}$$

(6) **Unavailability:**

$$(\neg x_{k,s,d}), \quad \forall k, s, d \text{ where } P_{k,s,d} = -1 \tag{6}$$

2.2.2 Soft Clauses.

(7) **Preferences:** To minimize the total cost of the schedule, we add a soft clause for each potential assignment that, if violated (i.e., if the nurse is assigned), adds its cost to the objective function.

$$(\neg x_{k,s,d}) \text{ with weight } P_{k,s,d}, \quad \forall k, s, d \text{ where } P_{k,s,d} \geq 0 \tag{7}$$

3 Algorithm and Configuration

The encoding was implemented in Python, and the resulting formula was provided to the RC2 MaxSAT solver [2] available in the PySAT library. The CaDiCaL 1.9.5 solver [1] was used as the backend oracle. We chose CaDiCal 1.9.5 simply because it far outperformed every other option we tried. All cardinality constraints were encoded using the Modulo Totalizer encoding [3], which requires $O(n^{3/2})$ clauses with $O(n \cdot log(n))$ variables.

To know the number of clauses and variables we need we simply sum up all the necessary components from each constraint:

- Initially we create nDS variables to represent all the $x_{k,s,d}$ boolean identification variables.
- As constraints (1), (2), (4), and (5) are all cardinality constraints, with (4) including 2 different cardinality constraints using the Modulo Totalizer encoding, we get $5 \cdot O((nDS)^{3/2})$ hard clauses and $5 \cdot O(nDS \cdot \log(nDS))$ auxiliary variables from these constraints.
- For constraint (3), we add $O(nDS \cdot \log(S))$ hard clauses: $O(nD(S-1) \cdot \log(S)) \Leftrightarrow O(nDS \cdot \log(S))$.
- For constraint (6) we add $O(nDS)$ hard clauses.
- For the minimization constraint (7) we add $O(nDS)$ soft clauses.

In total we create:

- nDS identification variables.
- $O(nDS \cdot \log(nDS))$ auxiliary variables from 5 cardinality constraints using the Modulo Totalizer encoding.
- For the hard clauses, the complexity is: $5 \cdot O((nDS)^{3/2}) + O(nDS \cdot \log(nDS)) + O(nDS)$. The dominant term is typically $O((nDS)^{3/2})$.
- $O(nDS)$ soft clauses, from the minimization constraint.

4 Installing and running

This project was implemented in Python. The sole dependency is PySAT, which can be installed via pip:

```
pip3 install pysat
```

The program can be executed with the following command:

```
python3 project1.py < [input.txt] > [output.txt]
```

References

[1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, Nils Froleyks, and Florian Pollitt. 2024. CaDiCaL 2.0. In *Computer Aided Verification - 36th International Conference, CAV 2024, Montreal, QC, Canada, July 24-27, 2024, Proceedings, Part I (Lecture Notes in Computer Science, Vol. 14681)*, Arie Gurfinkel and Vijay Ganesh (Eds.). Springer, 133–152. https://doi.org/10.1007/978-3-031-65627-9_7

[2] Alexey Ignatiev, Antonio Morgado, and Joao Marques-Silva. 2019. RC2: an Efficient MaxSAT Solver. *Journal on Satisfiability, Boolean Modelling and Computation* 11, 1 (2019), 53–64. https://doi.org/10.3233/SAT190116

[3] Toru Ogawa, Yangyang Liu, Ryuzo Hasegawa, Miyuki Koshimura, and Hiroshi Fujita. 2013. Modulo Based CNF Encoding of Cardinality Constraints and Its Application to MaxSAT Solvers. In *2013 IEEE 25th International Conference on Tools with Artificial Intelligence*. 9–17. https://doi.org/10.1109/ICTAI.2013.13